

Renormalization in Massive ϕ^4 Theory

Perimeter Scholars International, Quantum Field Theory II, Homework 2; 15 worked parts.

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Quantum Field Theory

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

1. Renormalization for Massive ϕ^4 Theory

Work in Euclidean signature and retain terms through one loop. Define the renormalized action by

$$S_R[\phi] = S[\phi] + \hbar \Delta_1 S[\phi] + O(\hbar^2), \quad (1.1)$$

where

$$S[\phi] = \int d^4x \left[\frac{1}{2}(\partial\phi)^2 + \frac{m_R^2}{2}\phi^2 + \frac{g_R}{4!}\phi^4 \right] \quad (1.2)$$

and

$$\Delta_1 S[\phi] = \int d^4x \left[\frac{B_1}{2}\phi^2 + \frac{C_1}{4!}\phi^4 \right]. \quad (1.3)$$

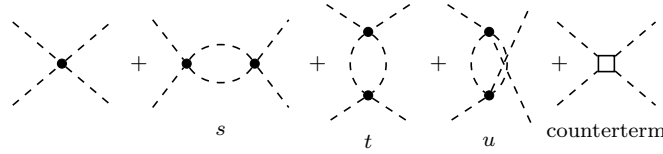
The cutoff-dependent coefficients B_1 and C_1 are chosen so that physical quantities expressed in terms of m_R and g_R remain finite as $\Lambda \rightarrow \infty$.

The one-loop four-point 1PI function is

$$\widehat{\Gamma}_R^{(4)}(p_i) \quad (1.4)$$

$$= g_R - \frac{\hbar g_R^2}{2} \left[I(p_1 + p_2, m_R) + I(p_1 + p_3, m_R) + I(p_1 + p_4, m_R) \right] + \hbar C_1 + O(\hbar^2). \quad (1.5)$$

Using a hollow square for the quartic counterterm, these contributions have the following topologies:



a) In four-dimensional Euclidean spacetime, write the sharp-cutoff integral corresponding to

$$I(p_1 + p_2, m_R; \Lambda), \quad (1.6)$$

where Λ regulates the ultraviolet divergence.

Solution.

$$\begin{aligned} I(p_1 + p_2, m_R; \Lambda) &= \int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 + m_R^2)[(k + p_1 + p_2)^2 + m_R^2]} \\ &= \frac{4\pi}{(2\pi)^4} \int_0^\pi d\chi \sin^2 \chi \int_0^\Lambda dk \frac{k^3}{(k^2 + m_R^2)(k^2 + s + 2k\sqrt{s} \cos \chi + m_R^2)}. \end{aligned} \quad (1.7)$$

b) Set $m_R^2 = 0$ and calculate the leading large- Λ behavior of $I(p, 0; \Lambda)$.

Solution.

$$\begin{aligned}
I(p_1 + p_2, 0; \Lambda) &= \frac{1}{4\pi^3} \int_0^\Lambda dk k \int_0^\pi d\chi \frac{\sin^2 \chi}{k^2 + s + 2k\sqrt{s} \cos \chi} \\
&= \frac{1}{4\pi^3} \int_0^\Lambda dk k \frac{\pi}{4k^2 s} \left(k^2 + s - \sqrt{(k^2 + s)^2 - 4k^2 s} \right) \\
&= \frac{1}{16\pi^2 s} \int_0^\Lambda \frac{1}{k} (k^2 + s - |k^2 - s|) \\
&= \frac{1}{16\pi^2 s} \left(\int_{\sqrt{s}}^\Lambda dk \frac{2s}{k} + \int_0^{\sqrt{s}} dk \frac{2k^2}{k} \right) \\
&= \frac{1}{16\pi^2 s} \left(1 + \log \frac{\Lambda^2}{s} \right). \tag{1.8}
\end{aligned}$$

c) By differentiating under the integral, show that

$$\frac{\partial}{\partial(m_R^2)} I(p, m_R; \Lambda) \tag{1.9}$$

is finite as $\Lambda \rightarrow \infty$. Conclude that the divergent part in the massive theory is the same as the one identified in the massless theory.

Solution. By differentiating with respect to m_R^2 , the integral now looks like

$$\sim \int_0^\Lambda dk \frac{k^3}{k^6}, \tag{1.10}$$

which is finite. Since the rate of change of the integral remains finite, the divergent part must be the same for massive and massless theory.

For parts d)–h), work in massless ϕ^4 theory.

d) Determine the divergent part of C_1 required to make $\hat{\Gamma}_R^{(4)}(p_i)$ finite and impose

$$\hat{\Gamma}_R^{(4)}(p_i^{\text{ref}}) = g_R, \quad |p_1^{\text{ref}} + p_2^{\text{ref}}| = |p_1^{\text{ref}} + p_3^{\text{ref}}| = |p_1^{\text{ref}} + p_4^{\text{ref}}| = \mu. \tag{1.11}$$

Only the terms in the counterterm that diverge as $\Lambda \rightarrow \infty$ need be given explicitly.

Solution. Using the result from b), we have

$$g_R - \frac{\hbar g_R^2}{2} \frac{3}{16\pi^2} \left(1 + \log \frac{\Lambda^2}{\mu^2} \right) + \hbar C_1 = g_R \implies C_1 = \frac{3g_R^2}{32\pi^2} \left(1 + \log \frac{\Lambda^2}{\mu^2} \right). \tag{1.12}$$

e) Find $\hat{\Gamma}_R^{(4)}(p_i, g_R, \mu)$ at general external momenta. Renormalization-scale independence requires

$$\hat{\Gamma}_R^{(4)}(p_i, g_R, \mu) = \hat{\Gamma}_R^{(4)}(p_i, g'_R, \mu'), \tag{1.13}$$

where the primed parameters are defined at the scale μ' .

Solution.

$$\begin{aligned}
\hat{\Gamma}_R^{(4)}(p_i, g_R, \mu) &= g_R - \frac{\hbar g_R^2}{32\pi^2} \left(\log \frac{\mu^2}{s} + \log \frac{\mu^2}{t} + \log \frac{\mu^2}{u} \right) \\
&= g'_R - \frac{\hbar g_R'^2}{32\pi^2} \left(\log \frac{\mu'^2}{s} + \log \frac{\mu'^2}{t} + \log \frac{\mu'^2}{u} \right). \tag{1.14}
\end{aligned}$$

f) Evaluate the preceding equality at a convenient momentum configuration and derive $g'_R(\mu')$ in terms of $g_R(\mu)$, μ , and μ' .

Solution. Take $s = u = t = \mu'^2$, we find

$$g'_R(\mu') = g_R(\mu) - \frac{3\hbar g_R^2(\mu)}{32\pi^2} \log \frac{\mu^2}{\mu'^2}. \quad (1.15)$$

g) Define the effective coupling by

$$g_{\text{eff}}(E) = \begin{cases} g_R(\mu), & E = \mu, \\ g'_R(\mu'), & E = \mu', \end{cases} \quad (1.16)$$

and the beta function by

$$\beta(g_{\text{eff}}) \equiv E \frac{d}{dE} g_{\text{eff}}(E). \quad (1.17)$$

Compute the one-loop beta function and plot it as a function of g_{eff} .

Solution. The definition here is not very conventional. It is equivalent to defining

$$g_{\text{eff}}(E) = \Gamma_R^{(4)}(E, g_R, \mu), \quad (1.18)$$

which immediately implies

$$g_{\text{eff}}(E) = \begin{cases} g_R(\mu), & E = \mu, \\ g'_R(\mu'), & E = \mu'. \end{cases} \quad (1.19)$$

Then, we have

$$\beta(g_{\text{eff}}) = \frac{3\hbar g_R^2(\mu)}{16\pi^2} = \frac{3\hbar g_{\text{eff}}^2}{16\pi^2}. \quad (1.20)$$

h) Explain what this beta function implies for the interaction strength at high and low energies.

Solution. Solving the beta function, we have

$$\frac{1}{g_{\text{eff}}(E)} = \frac{1}{g_0} + \frac{3\hbar}{32\pi^2} \log \frac{\Lambda^2}{E^2}. \quad (1.21)$$

Therefore, the theory is free at IR and becomes strongly coupled at the UV.

The four-point diagrams introduce no additional divergence in the massive theory, so the same calculation can be repeated there. For mass renormalization, consider the one-loop two-point 1PI function

$$\hat{\Gamma}_R^{(2)}(p) = p^2 + m_R^2 + \frac{\hbar g_R}{2} T(m_R) + \hbar B + O(\hbar^2). \quad (1.22)$$

Using a filled square for the quadratic counterterm, its topologies are

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i) Write the sharp-cut-off integral $T(m_R; \Lambda)$ that regulates the UV divergences.

Solution.

$$T(m_R; \Lambda) = \int \frac{d^4 k}{(2\pi)^4} \frac{1}{k^2 + m^2} = \frac{1}{8\pi^2} \int_0^\Lambda dk \frac{k^3}{k^2 + m^2}. \quad (1.23)$$

j) Evaluate $T(m_R; \Lambda)$ at large Λ . Identify separately the quadratic divergence, logarithmic divergence, and term that remains finite as $\Lambda \rightarrow \infty$.

Hint: Radial coordinates may be useful. The area of the unit n -sphere is

$$\int d^n \Omega = \frac{2\pi^{n/2}}{\Gamma(n/2)}. \quad (1.24)$$

Solution.

$$T(m_R; \Lambda) = \frac{1}{16\pi^2} \left(\Lambda^2 - m_R^2 \log \frac{\Lambda^2 + m_R^2}{m_R^2} \right) = \frac{1}{16\pi^2} \left(\Lambda^2 - m_R^2 \log \frac{\Lambda^2}{m_R^2} + \mathcal{O}(\Lambda^{-2}) \right). \quad (1.25)$$

The physical mass is determined by the pole of the propagator, equivalently by

$$\hat{\Gamma}_R^{(2)}(p_i^{\text{ref}}, m_R, g_R, \mu) = 0, \quad |p_i^{\text{ref}}| = m_{\text{phys}}, \quad (1.26)$$

for any μ . For practical subtraction, impose instead

$$\hat{\Gamma}_R^{(2)}(0, m_R, g_R, \mu) = m_R^2, \quad \mu = m_R, \quad (1.27)$$

and take the mass counterterm to have the form

$$B(m_R, g_R, \mu, \Lambda) = B_0(g_R, \mu, \Lambda) + m_R^2 B_1(g_R, \mu, \Lambda). \quad (1.28)$$

k) Determine $B_0(g_R, \mu, \Lambda)$ and $B_1(g_R, \mu, \Lambda)$ so that the preceding renormalization condition holds.

Solution. We can rewrite

$$\hat{\Gamma}_R^{(2)}(p, m_R, g_R, \mu) = p^2 + m_R^2 + \hbar \left(\frac{g_R(\mu)\Lambda^2}{32\pi^2} + B_0 + m_R^2 \left(-\frac{g_R(\mu)}{32\pi^2} \log \frac{\Lambda^2}{\mu^2} - \frac{g_R(\mu)}{32\pi^2} \log \frac{\mu^2}{m_R^2} + B_1 \right) \right). \quad (1.29)$$

The renormalisation condition implies

$$B_1(g_R, \mu, \Lambda) = \frac{g_R}{32\pi^2} \log \frac{\Lambda^2}{\mu^2}, \quad B_0(g_R, \mu, \Lambda) = -\frac{g_R \Lambda^2}{32\pi^2}. \quad (1.30)$$

l) Check that these counterterms make $\hat{\Gamma}_R^{(2)}(p_i, m_R, g_R, \mu)$ finite for arbitrary m_R and p .

Solution.

$$\hat{\Gamma}_R^{(2)}(p, m_R, g_R, \mu) = p^2 + m_R(\mu)^2 - \frac{\hbar g_R(\mu)}{32\pi^2} \log \frac{\mu^2}{m_R(\mu)^2}. \quad (1.31)$$

m) Require $\hat{\Gamma}_R^{(2)}$ to be independent of the renormalization scale, as for the four-point function, and derive an equation relating $[m'_R(\mu')]^2$ and $[m_R(\mu)]^2$. Retain terms only through order $\hbar$ and express the result in terms of g_R .

Solution.

$$m_R^2(\mu) - \frac{\hbar g_R(\mu) m_R^2(\mu)}{32\pi^2} \log \left(\frac{\mu^2}{m_R^2(\mu)} \right) = [m'_R(\mu')]^2 - \frac{\hbar g'_R(\mu') [m'_R(\mu')]^2}{32\pi^2} \log \left(\frac{\mu'^2}{[m'_R(\mu')]^2} \right). \quad (1.32)$$

This implies

$$[m'_R(\mu')]^2 = m_R^2(\mu) + \frac{\hbar g_R(\mu) m_R^2(\mu)}{32\pi^2} \log \left(\frac{\mu'^2}{\mu^2} \right) + \mathcal{O}(\hbar^2). \quad (1.33)$$

n) Show that when $\mu^2 = m_R^2$, the renormalized mass agrees in magnitude with the physical mass:

$$|m_R| = |m_{\text{phys}}|. \quad (1.34)$$

Solution.

$$\hat{\Gamma}_R^{(2)}(|m_{\text{phys}}|, m_R = \mu, g_R, \mu) = 0 \implies |m_{\text{phys}}| = |m_R|. \quad (1.35)$$

o) Calculate the anomalous dimension of the mass-squared operator,

$$\gamma_{m^2}(g_R) \equiv \mu \frac{\partial}{\partial \mu} \ln m_R^2(\mu). \quad (1.36)$$

Solution. Starting from the physical mass

$$m_{\text{phys}}^2 = m_R^2(\mu) \left[1 - \frac{\hbar g_R(\mu)}{32\pi^2} \log \left(\frac{\mu^2}{m_R^2(\mu)} \right) \right] + O(\hbar^2), \quad (1.37)$$

and hence

$$\log m_{\text{phys}}^2 = \log m_R^2(\mu) - \frac{\hbar g_R(\mu)}{32\pi^2} \log \left(\frac{\mu^2}{m_R^2(\mu)} \right) + O(\hbar^2). \quad (1.38)$$

Then,

$$0 = \frac{d \log m_{\text{phys}}^2}{d \log \mu} = \gamma_{m^2}(g_R) - \frac{2\hbar g_R}{32\pi^2} + O(\hbar^2), \quad (1.39)$$

which implies

$$\gamma_{m^2}(g_R) = \frac{\hbar g_R}{16\pi^2} + O(\hbar^2). \quad (1.40)$$

At one loop in ϕ^4 theory, no wavefunction renormalization is required; the two-point correction is momentum independent at this order.

2. Renormalization Group Flow for the $O(N)$ Model

Let $\vec{\phi} = \{\phi_\alpha\}_{\alpha=1}^N$ be an N -component scalar field with $O(N)$ symmetry. In D -dimensional Euclidean space, take

$$S_E[\vec{\phi}] = \int d^D x \left[\frac{1}{2} (\nabla \vec{\phi})^2 + U(\vec{\phi}^2) \right], \quad (2.1)$$

where

$$X \equiv \vec{\phi}^2 = \sum_{\alpha=1}^N \phi_\alpha^2, \quad (2.2)$$

and $U(X)$ is regular. Restrict the Fourier support to $|k| \leq \Lambda$ and write

$$Z = \int \mathcal{D}_\Lambda[\vec{\phi}] \exp \left[-\frac{1}{\hbar} S_E[\vec{\phi}] \right]. \quad (2.3)$$

Separate the field into slow and fast components,

$$\vec{\phi} = \tilde{\vec{\phi}} + \vec{\phi}_{\text{fast}}, \quad (2.4)$$

where $\tilde{\vec{\phi}}$ has modes $|k| \leq \Lambda/S$ and $\vec{\phi}_{\text{fast}}$ has modes $\Lambda/S < |k| \leq \Lambda$. Integrate a thin shell using

$$S = 1 + ds, \quad 0 < ds \ll 1. \quad (2.5)$$

- Hold $\tilde{\vec{\phi}}$ fixed and expand $U(X)$ through second order in $\vec{\phi}_{\text{fast}}$ (the Gaussian approximation), expressing the result in terms of U' and U'' . Keep the vector structure explicit.
- In the local-potential approximation, take $\tilde{\vec{\phi}}$ to be constant, so only its $k=0$ mode is nonzero. Show that the fast-mode integral separates into one longitudinal mode parallel to $\tilde{\vec{\phi}}$ and $N-1$ decoupled transverse modes.
- Show that the two kinds of mode renormalize the potential according to

$$U \longrightarrow U_{\text{eff}} = U + ds (\Delta U_{\parallel} + (N-1)\Delta U_{\perp}), \quad (2.6)$$

and calculate $\Delta U_{\parallel}(X)$ and $\Delta U_{\perp}(X)$.

d) Show that the rescalings

$$\vec{\phi} \longrightarrow (1 - ds)^{(2-D)/2} \vec{\phi}, \quad x \longrightarrow (1 - ds)x \quad (2.7)$$

restore the cutoff to Λ .

e) Determine the local-potential flow equation

$$S \frac{\partial}{\partial S} U_S(X) = ? \quad (2.8)$$

for the $O(N)$ model. As in the lecture, set $\Lambda = 1$ if desired.

f) Check that $N = 1$ reproduces the one-component flow for $V(\phi)$, using $U(\phi^2) = V(\phi)$.

g) Truncate the potential at second order in $X = \vec{\phi}^2$,

$$U(X) = \frac{t}{2}X + \frac{u}{8}X^2. \quad (2.9)$$

Expand the flow equation consistently through X^2 . Derive both the flow equations and the Wilson functions $S \partial g_S / \partial S$ for the renormalized couplings t_S and u_S .